CONNECTEDNESS PLUS A POINT WITH ALGEBRAICALLY CLOSED RESIDUE FIELD GIVES GEOMETRIC CONNECTEDNESS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the characterization applied to X; no step is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-connected-if-connected-and-point is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- A field extension in which k is algebraically closed becomes geometrically irreducible after base change. Algebra, Lemma algebra-lemma-field-extension-geometrically-irreducible, invoked at proof lines 1093-1095. Supplies irreducibility of the base change of the spectrum of the residue field.
- Criterion for geometric connectedness through a connected morphism to an irreducible base. Same chapter, Lemma lemma-geometrically-connected-criterion, invoked at proof lines 1096-1098. Supplies connectedness of X over the separable closure.
- Characterization of geometric connectedness by connectedness over the separable closure. Same chapter, Lemma lemma-characterize-geometrically-connected, invoked at proof lines 1098-1100. Supplies the final conclusion.

Source reference. Original lemma 037P: sources/stacks-chow/tags.txt, source lines 4208–4208. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0387: sources/stacks-varieties/pinned-original-source.tex, source lines 792–792. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 056R: sources/stacks-varieties/pinned-original-source.tex, source lines 1067–1067. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Assume X is connected and has a point x such that k is algebraically closed in $\kappa(x)$. Then X is geometrically connected. In particular, if X has a k-rational point and X is connected, then X is geometrically connected.*

Proof. Set $T = \mathrm{Spec}(\kappa(x))$. Let $\overline{k}$ be a separable algebraic closure of k. The assumption on $\kappa(x)/k$ implies that $T_{\overline{k}}$ is irreducible, see Algebra, Lemma 037P. Hence by Lemma 056R we see that $X_{\overline{k}}$ is connected. By Lemma 0387 we conclude that X is geometrically connected. □